

PSWF-Based Regularization of a One-Dimensional Bandlimited Inverse Problem

1 Overview and Aim

Prolate spheroidal wave functions (PSWFs) arise naturally in the classical problem of simultaneous time and frequency concentration [1, 2, 3]. They are particularly relevant to the present inverse problem because they are eigenfunctions of both the finite Fourier transform and the associated sinc-kernel concentration operator. Consequently, the forward operator has a diagonal representation in the PSWF basis.

The purpose of this numerical project is not merely to reconstruct a smooth test function. Its main goal is to make three features of a compact inverse problem visible in a single reproducible MATLAB experiment:

1. the transition from well-determined to poorly determined spectral modes;
2. the amplification of noise produced by direct inversion;
3. the stabilization obtained by spectral cutoff and Tikhonov regularization.

A second goal is to explain what is gained by using PSWFs instead of an arbitrary basis. Since the PSWF basis diagonalizes the forward operator, the inverse problem reduces to scalar operations on its concentration eigenvalues. The code therefore compares the representation of the same operator in a PSWF basis and in a normalized Legendre basis.

The experiment uses the bandwidth parameter

$$c = 10.$$

This is intentionally larger than in the introductory example. It produces a sizeable collection of significant eigenvalues and makes the spectral transition much easier to study.

2 The Forward Operator and the Role of c

On $L^2([-1, 1])$, consider

$$(K_c f)(x) = \int_{-1}^1 \frac{\sin(2\pi c(x-t))}{\pi(x-t)} f(t) dt. \quad (1)$$

At $x = t$, the kernel has a removable singularity and its continuous value is

$$\lim_{t \rightarrow x} \frac{\sin(2\pi c(x-t))}{\pi(x-t)} = 2c.$$

This is the reason that the diagonal entries of the discretized kernel are set equal to $2c$ in the MATLAB code.

Let

$$K_c \psi_n = \lambda_n \psi_n, \quad \|\psi_n\|_{L^2(-1,1)} = 1.$$

For this normalization the concentration eigenvalues satisfy the pointwise spectrum-accumulation identity

$$\sum_{n=0}^{\infty} \lambda_n |\psi_n(x)|^2 = 2c. \quad (2)$$

Integrating (2) over $[-1, 1]$ and using the normalization of the PSWFs gives

$$\sum_{n=0}^{\infty} \lambda_n = 4c. \quad (3)$$

Thus, when $c = 10$, the total spectral mass is 40. Since the concentration eigenvalues lie between zero and one and exhibit the well-known Slepian transition, the index

$$N_{\text{Shannon}} \approx 4c = 40$$

provides an effective dimension for this experiment. This does not mean that exactly forty eigenvalues are equal to one. Rather, many eigenvalues are close to one, followed by a relatively narrow transition region and then a rapidly decaying tail.

For this reason the MATLAB implementation computes 55 PSWF modes. This is enough to include the well-concentrated modes, the transition region, and several eigenvalues that are so small that direct inversion becomes numerically dangerous.

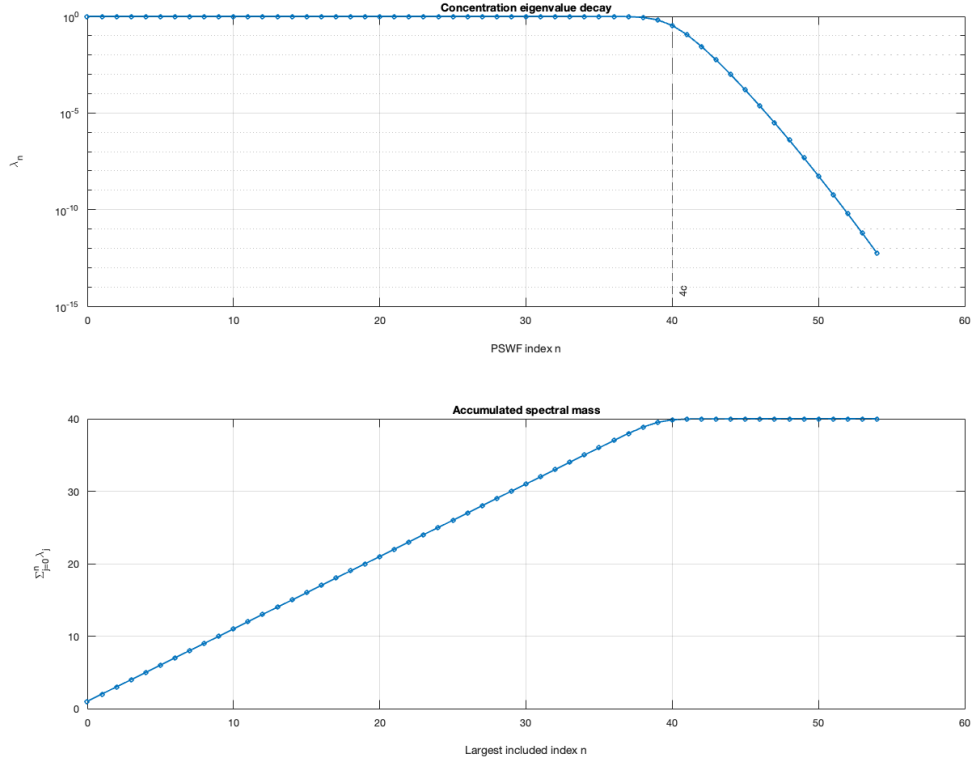


Figure 1: PSWF concentration spectrum for $c = 10$. The upper panel shows the decay of λ_n , with $4c = 40$ indicated as the effective time–bandwidth dimension. The lower panel shows the accumulated spectral mass $\sum_{j=0}^n \lambda_j$, which approaches the trace value $4c$.

3 A More Structured Test Signal

To force the reconstruction to use substantially more than a few low-order modes, the test signal contains several localized components together with oscillatory terms:

$$f(x) = 0.95e^{-110(x+0.58)^2} - 0.75e^{-75(x+0.12)^2} + 0.90e^{-95(x-0.42)^2} + 0.24\cos(12\pi x + 5x^2) + 0.16\sin(18\pi x) + 0.10x. \quad (4)$$

The synthetic noisy data are

$$g^\delta = K_c f + \eta, \quad (5)$$

where the noise is scaled so that

$$\frac{\|\eta\|_2}{\|K_c f\|_2} = 5 \times 10^{-4}.$$

The combination of localized peaks and oscillatory components makes the example less forgiving than a single Gaussian. At the same time, it remains simple enough that the behavior of the inverse methods can be interpreted directly.

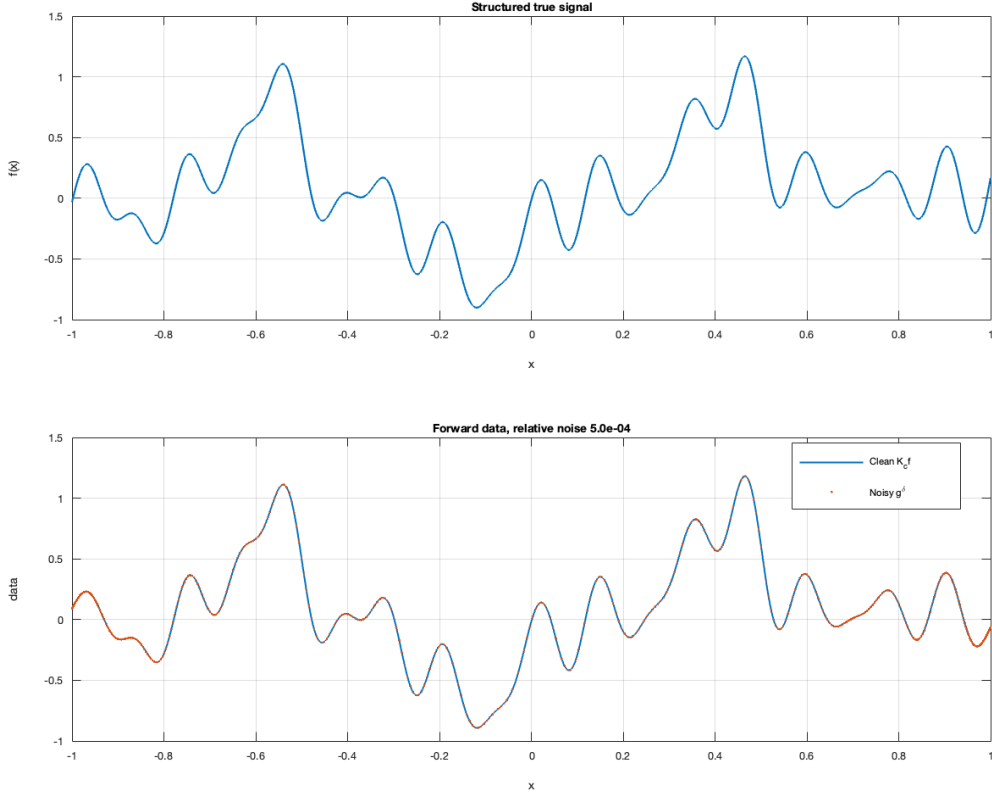


Figure 2: Synthetic forward experiment for $c = 10$. The upper panel shows the structured test signal in (4). The lower panel compares the exact data $K_c f$ with the noisy measurements g^δ .

4 Why Direct Inversion Is Unstable

Expanding

$$f = \sum_{n=0}^{\infty} a_n \psi_n$$

and using $K_c \psi_n = \lambda_n \psi_n$ gives

$$K_c f = \sum_{n=0}^{\infty} \lambda_n a_n \psi_n.$$

For exact data,

$$a_n = \frac{\langle K_c f, \psi_n \rangle}{\lambda_n}.$$

For noisy data,

$$\frac{\langle g^\delta, \psi_n \rangle}{\lambda_n} = a_n + \frac{\langle \eta, \psi_n \rangle}{\lambda_n}.$$

The second term explains the instability. Once λ_n is small, even a modest noise coefficient can become very large after division by λ_n . This is the spectral form of the standard Picard description of a compact ill-posed inverse problem [7, 6].

The code deliberately computes the naive inverse using all retained modes. It is not included as a sensible reconstruction method; it is included to show what goes wrong when the small eigenvalues are inverted without regularization.

5 Spectral Cutoff Regularization

A direct way to suppress the unstable tail is to retain only the first k PSWF modes:

$$f_k^\delta = \sum_{n=0}^{k-1} \frac{\langle g^\delta, \psi_n \rangle}{\lambda_n} \psi_n. \quad (6)$$

This is the continuous PSWF counterpart of truncated singular-value decomposition (TSVD).

If k is too small, recoverable information is discarded. If k is too large, noise-dominated coefficients associated with small λ_n enter the solution. The MATLAB experiment therefore records both the reconstruction error and the data residual for every cutoff.

6 Tikhonov Regularization

Tikhonov regularization replaces the unstable inverse by the minimization problem

$$f_\alpha^\delta = \arg \min_f \left\{ \|K_c f - g^\delta\|_2^2 + \alpha \|f\|_2^2 \right\}, \quad \alpha > 0. \quad (7)$$

This is the standard zero-order Tikhonov functional [4, 6].

Because the PSWFs diagonalize K_c , the minimization separates into scalar problems. If

$$b_n^\delta = \langle g^\delta, \psi_n \rangle,$$

then

$$f_\alpha^\delta = \sum_{n=0}^{\infty} \frac{\lambda_n}{\lambda_n^2 + \alpha} b_n^\delta \psi_n. \quad (8)$$

The filter

$$q_\alpha(\lambda) = \frac{\lambda}{\lambda^2 + \alpha}$$

behaves approximately like $1/\lambda$ when $\lambda^2 \gg \alpha$, so the well-determined modes are almost unchanged. When $\lambda^2 \ll \alpha$, however,

$$q_\alpha(\lambda) \approx \frac{\lambda}{\alpha},$$

and the dangerous division by a tiny eigenvalue is suppressed.

The central practical question is therefore not whether to use α , but how to choose it.

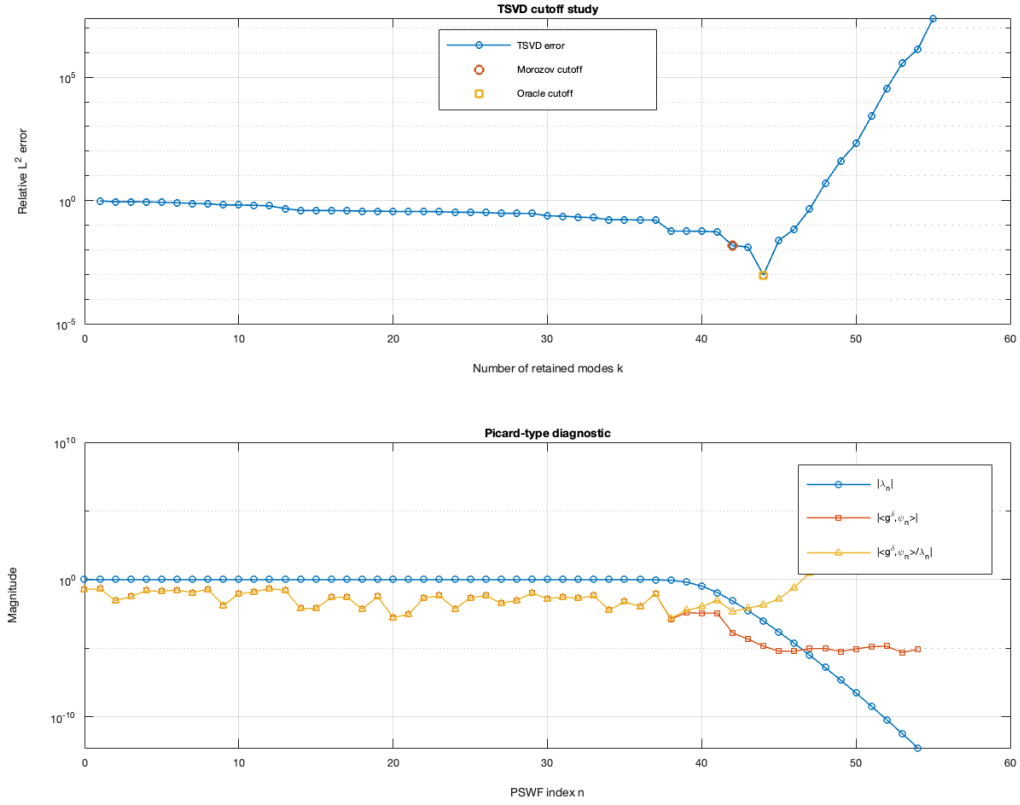


Figure 3: Spectral-cutoff and Picard diagnostics. The upper panel shows the reconstruction error as the number of retained PSWF modes changes. The lower panel displays λ_n , the noisy data coefficients $|\langle g^\delta, \psi_n \rangle|$, and the formally inverted coefficients. Their growth after the spectral transition illustrates the onset of noise amplification.

6.1 Choosing α : Morozov's discrepancy principle

Suppose that an estimate of the absolute noise level is available,

$$\|\eta\|_2 \leq \delta.$$

Morozov's discrepancy principle chooses the regularization parameter so that the regularized solution fits the measured data only to the accuracy justified by the noise:

$$\|K_c f_\alpha^\delta - g^\delta\|_2 \approx \tau \delta, \quad \tau > 1. \quad (9)$$

The logic is important. A residual much larger than δ indicates that the solution has been over-regularized and does not explain the data well enough. A residual far below the noise level attempts to fit features that cannot be distinguished reliably from the noise and therefore tends toward overfitting.

In the MATLAB experiment,

$$\tau = 1.05.$$

The code evaluates the Tikhonov solution on a logarithmic grid of candidate values of α and selects

$$\alpha_{\text{DP}} = \arg \min_{\alpha} \left| \|K_c f_\alpha^\delta - g^\delta\|_2 - \tau \delta \right|. \quad (10)$$

This parameter can be selected without knowing the exact solution f .

Because the present problem is synthetic, the code also computes an *oracle* value

$$\alpha_{\text{oracle}} = \arg \min_{\alpha} \|f_\alpha^\delta - f\|_2.$$

This second value is useful only for benchmarking: it cannot be used with real data because the true f is precisely the unknown quantity. The comparison between α_{DP} and α_{oracle} therefore tells us how much accuracy is lost when a realizable parameter-choice rule is used.

Morozov's discrepancy principle is a classical a-posteriori parameter choice for regularization of ill-posed problems [5, 6, 7]. When the noise norm is not known, alternatives include generalized cross-validation and L-curve criteria.

7 Reconstruction Results

The four reconstructions are displayed separately so that the very large amplitude of the unstable naive inverse does not hide the regularized solutions. The two practical reconstructions shown in Figure 5 use Morozov's discrepancy principle: one to select a spectral cutoff and one to select the Tikhonov parameter.

8 Why Use PSWFs?

The main advantage of PSWFs here is not that they constitute a universally better approximation basis. Their advantage is that they are *adapted to this particular forward operator*.

8.1 The forward operator is diagonal

In the PSWF basis,

$$\langle K_c \psi_n, \psi_m \rangle = \lambda_n \delta_{nm}.$$

Thus the forward map, the inverse instability, TSVD, and Tikhonov regularization can all be understood one mode at a time. No numerical SVD of the discretized sinc matrix is required once the PSWF family has been constructed.

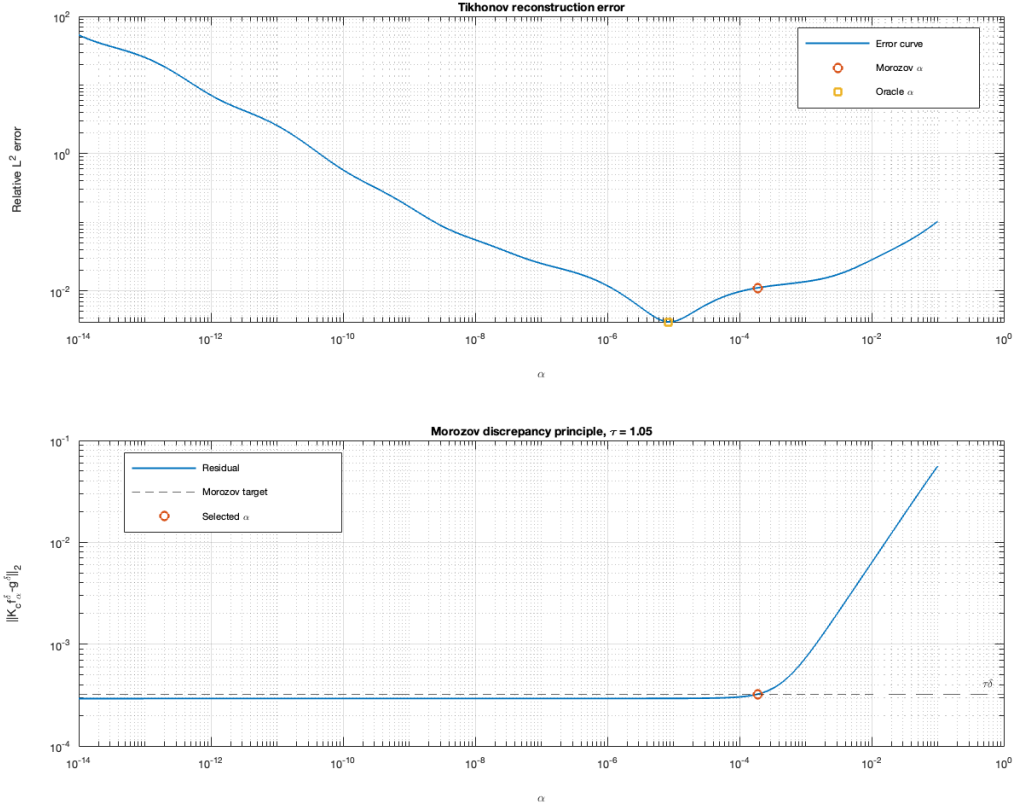


Figure 4: Selection of the Tikhonov parameter. The upper panel shows the true reconstruction error, which is available only in this synthetic test and therefore supplies the oracle value. The lower panel shows the residual $\|K_{\alpha} f_{\alpha}^{\delta} - g^{\delta}\|_2$ and the Morozov target $\tau\delta$. The practical value α_{DP} is chosen from the intersection of these quantities.

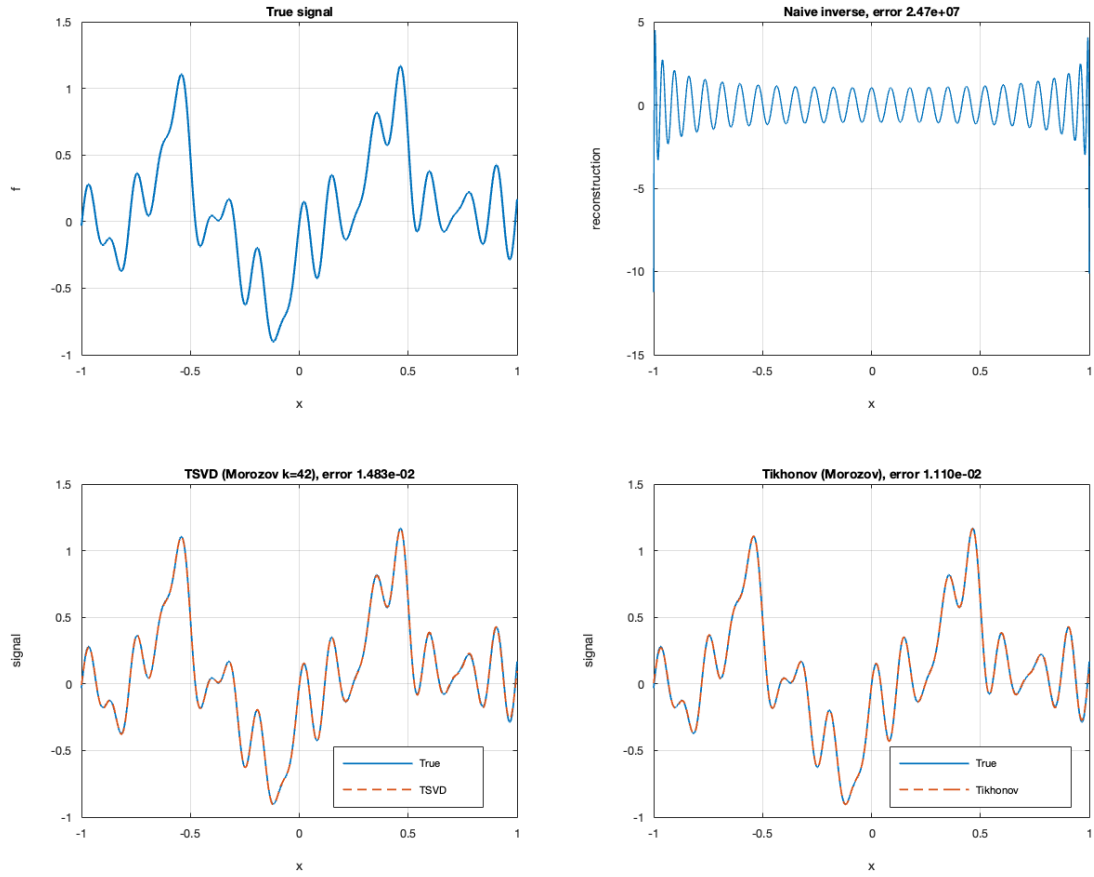


Figure 5: Comparison of the true signal, the unregularized inverse, the PSWF spectral-cutoff reconstruction, and the Tikhonov reconstruction. The regularized solutions use parameters selected by the discrepancy principle, whereas the naive inverse retains all computed modes.

In contrast, a generic orthogonal basis such as normalized Legendre polynomials does not diagonalize K_c . The MATLAB code explicitly forms the two matrices

$$A_{mn}^{\text{PSWF}} = \langle K_c \psi_n, \psi_m \rangle$$

and

$$A_{mn}^{\text{Leg}} = \langle K_c \bar{P}_n, \bar{P}_m \rangle,$$

and compares their off-diagonal energy.

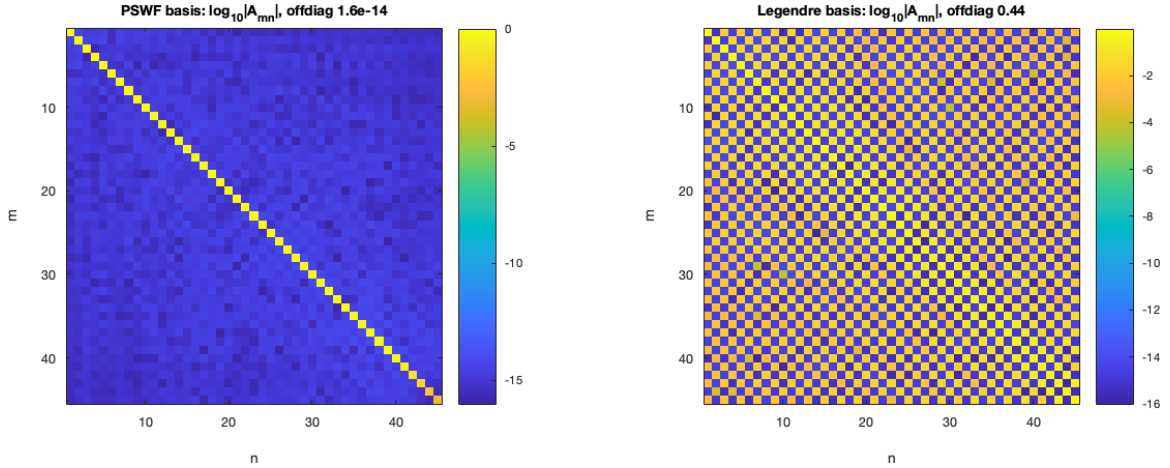


Figure 6: Representation of the same sinc-kernel operator in two orthonormal bases. In the PSWF basis the matrix is essentially diagonal; in the normalized Legendre basis substantial off-diagonal coupling remains. This illustrates the operator-adapted nature of PSWFs.

8.2 The eigenvalues have a direct information interpretation

The values λ_n measure the concentration of the corresponding bandlimited PSWF modes. Consequently, the same numbers that describe the classical time–frequency concentration problem also identify which components of the inverse problem are stably recoverable. The transition near the time–bandwidth dimension $4c$ therefore has an immediate inverse problem interpretation.

8.3 The basis belongs to the continuous problem

A singular-value decomposition of a sampled kernel matrix also reveals the ill-conditioning, and in a sufficiently accurate discretization its singular vectors approximate the PSWFs. The PSWF approach nevertheless has an important conceptual advantage: the basis comes from the continuous finite-Fourier/concentration problem and can be computed through the commuting Sturm–Liouville operator. It is therefore not tied to one particular sampling grid.

This operator-adapted structure is also why PSWFs continue to appear in reconstruction problems based on restricted Fourier data; see, for example, the one-dimensional PSWF reconstruction formulas and stability analysis in [8].

8.4 What PSWFs do not provide

The advantage is specific rather than universal. For an unrelated forward operator, PSWFs need not diagonalize the problem and may offer no benefit over an SVD, wavelets, splines, Fourier bases, or problem-specific regularizers such as total variation. Moreover, choosing a PSWF

basis does not remove ill-posedness: the rapidly decaying eigenvalues remain and regularization is still essential.

Thus the appropriate conclusion is not that PSWFs replace general inverse problem methods. Rather, for finite-Fourier and sinc-kernel problems they supply the natural spectral coordinates in which those methods become especially transparent.

9 Numerical Implementation

The code uses 500-point Gauss–Legendre quadrature rather than a uniform trapezoidal rule. This is useful when $c = 10$ because more PSWF modes are included and the high-order orthogonality relations must be resolved accurately. The PSWFs are constructed from the eigenvectors of the truncated differential-operator matrix and evaluated in the normalized Legendre basis

$$\bar{P}_k(x) = \sqrt{k + \frac{1}{2}} P_k(x).$$

Running

```
pswf_inverse_c10
```

prints the spectral diagnostics, the selected regularization parameters, and the reconstruction errors. It also creates the six figures used in this document in the directory `results/`.

10 Conclusion

This experiment gives a more substantial demonstration of a PSWF-based inverse problem than the low-bandwidth example. With $c = 10$, the effective spectral dimension is approximately $4c = 40$, so the computation contains a sizeable block of stable modes followed by a visible transition and a rapidly decaying eigenvalue tail. A structured signal with localized and oscillatory components requires many of those modes and makes the distinction between stable and unstable inversion clear.

The experiment also separates two issues that are sometimes conflated. PSWFs provide a basis that is exceptionally well matched to the finite-Fourier/sinc-kernel operator, but they do not themselves regularize the inverse problem. Stability still requires a decision about how much of the small-eigenvalue information should be trusted. Spectral cutoff makes that decision abruptly, whereas Tikhonov regularization makes it smoothly.

For the Tikhonov method, the practical parameter is selected here by Morozov’s discrepancy principle rather than by knowledge of the exact solution. The oracle parameter is retained only as a diagnostic benchmark. This distinction makes the numerical experiment closer to the way a real inverse problem would be treated.

Finally, the comparison with the Legendre basis makes the specific value of PSWFs visible: in PSWF coordinates the forward operator is essentially diagonal, while in a generic orthogonal basis the modes remain coupled. This gives a concrete computational interpretation to the classical time–frequency concentration theory and provides a natural starting point for later work on restricted Fourier data, nonuniform sampling, and multidimensional PSWF or Clifford-PSWF inverse problems.

References

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